Engineering Chemistry · Unit_03 · Energy storage devices, hydrogen energy, and sensors

DRAFT — UE26CY151A · Unit_03 · Energy storage devices, hydrogen energy, and sensors · Zinc -air battery

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Zinc -air battery

1 Zinc -air battery

Study focus: Zinc -air battery. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000070 · EV0000071

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module 5 on energy storage devices.

Key details

- That is, we are talking about batteries.
- For the Z care battery we look at the construction, working, advantages, disadvantages, and applications.
- So zinc air battery can be classified as a metal air battery because as you can see in the name the zinc and the air, so it is metal air batteries.
- Another important fact about this battery is that it's an alkaline battery.
- The most important thing about this battery is that it is using oxygen directly from the atmosphere to produce electrochemical energy.
- Property of this battery is that the energy density is going to be very high.
- The energy density for such batteries is very high.
- Battery has been considerably reduced.
- The reason being that one of the electroactive material that is the cathode electroactive material which is oxygen need not be stored inside the battery and can be taken directly from air.
- Salient feature in fact of this battery is that it has high energy density.
- For in a battery, in the first class when we talked about batteries, I told you there are four components, basic components of a battery, the anode,.
- The cathode, the electrolyte, and the separator.
- We have already studied just now that zinc granule zinc is the anode active material.
- In what form is it present?

- Zinc is taken in the form of granules, and why would that be to increase the surface area so that the reaction can happen at a faster rate.
- The role of this gelling agent is to hold the granules together.
- It is activity material is oxygen, but then we know that there needs to be a current collector.
- Just having the electroactive material is not sufficient.
- Yes, oxygen will be taken from the air, but we need to have electroactive material.
- electroactive material is taken from the air, but we need to have a current collector.
- Responsible for picking up the electrons or providing the electrons for reduction in this case because it is the cathode.
- We have carbon.
- That is in the form of crepite.
- That is because.
- At the cathode, oxygen has to undergo reduction, which is a slow process.
- In order to expedite the process, in order to make the reaction happen faster, we need to use a catalyst.

Example or application from the lecture

- Let us look at the construction of such a battery.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 05:00–09:59

Why would that be required?

Key details

- We are taking oxygen from the atmosphere.
- So therefore, in order to.
- Remove the moisture, not allow the moisture into the battery, we need to add a wetproofing agent, and all this is now coated on a nickel wire mesh support which is acting as the current collector.
- And apart from this, we also have an outer.
- layer of air permeable teflon layer.
- Now what would this be what would be the role of this teflon layer it's very obvious that teflon layer will pr prevent the moisture from entering into the battery, but at the same time it has to be air permeable also because it has to allow the oxygen to enter the battery.
- Though there is no cathode active material still there has to be a cathode inside the battery.
- Of course this cathode is going to form a very thin layer in the a thin small part of the battery and the remaining part will be the anode.
- through this air permeable Teflon layer, so we have air access holes on the cathode side, which are going to provide the pathway for oxygen to enter the battery.
- Why it has to be like this.
- The separator is soaked in electrolyte.
- We'll learn about it in a few minutes now.

- That is the construction.
- Positive that is the cathode.
- Through these holes.
- So this has to be cathode because oxygen undergoes reduction.
- the hydrophobic membrane that is your teflon layer, which we mentioned just now.
- Carbon and catalyst also to in order to for one carbon is for conductivity and catalyst is to speed up the reaction.
- Soaked in electrolyte.
- Separator soaked in electrolyte.
- So we can see that we because oxygen is being taken from the.
- Atmosphere directly, so it need not be stored inside the battery.
- Therefore, the entire battery space is actually filled with the anode active material and that is zinc.
- And then we have a cap here the metal cover which is the act acting as the negative terminal and we have a cup over here which is going to act as the positive terminal.
- The gasket is going to.
- Prevent the anode and cathode coming in contact with each other.
- At the same time, it is going to seal the two metal plates together, the metal cup and the metal cover together.
- This plastic gasket is placed in between, and therefore this.
- This is the all the components which we saw just now.
- They are all placed like this inside the battery.
- Then we have the anode terminal and the cathode terminal, and the in between there is a.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 10:00–14:59

gasket which is for preventing the anode and cathode from coming in direct contact with each other.

Key details

- Plus 2OH minus is giving you ZnO plus H2O plus 2 electrons.
- We should put the two electrons here because we are talking of the anodic reaction.
- Oxygen has to undergo reduction.
- The reaction is half O2 plus H2O plus 2 electron is giving you 2OH minus.
- Plus half O2 is giving you ZnO.
- Zinc is reacting with oxygen to give you Zn oxide.
- But if you look at the schematic diagram which when I talked about battery we I showed you the diagram where we talked about working of the battery we have the anode we have the cathode and we have the electrolyte and the separator.
- Is the anode material, and at the cathode we have this catalyst which is placed here, and we have this.
- Holes through which oxygen can enter.

- This is just a schematic representation and not the actual representation because the electrolyte is not.
- Present in bulk like this, but it is present in the separator only.
- The separator soaked in electrolyte.
- is produced here and this move towards the moves towards the zinc electrode and then we have this reaction happen at the animal.
- So electrons are being produced here, they are getting absorbed here and therefore we get power from this battery.
- Course batteries all batteries are galvanic cells, so this is how this the battery works.
- This is about the working of the battery, and we have.
- EMF is 1.4 volts.
- The separator soaked in electrolyte.
- The reason for that, if you look at the final reaction, zinc plus half O2 is giving you Z and O.
- OH minus is not figuring in the final reaction.
- That means the amount of OH minus used.
- is not going to affect the working of the battery.
- Whatever is produced at the cathode is being used at the anode.
- Just a small amount of QH just to facilitate the movement of the ions is what is required.
- Hence, we need not use a large amount of electrolyte and the separator soaked in electrolyte is sufficient, a small amount of electrolyte.
- Which is soaked inside the separator is enough for the battery to function and therefore.
- One important characteristic of this battery is that these batteries can be miniaturized.
- Because the amount of electrolyte is not important.
- If you look at lead acid battery, we know that in the final equation we have H2SO4.
- Of H2SO4 matters, and therefore, lead acid batteries always come in large sizes.

Example or application from the lecture

- Let us look at the working of this battery.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 15:00–19:59

You have to make them large, it cannot be made small.

Key details

- Whereas here OH minus the electrolyte is not there in the final equation.
- So that means final reaction.
- So amount of electrolyte can be very small, and therefore these batteries can be made into very small sizes.
- The word the term we use to describe that is that these batteries can be miniaturized.
- high energy density as I explained right in the beginning and that is because air which is the cathode active material is not getting stored inside the battery it is taken directly from the

atmosphere therefore the weight of the battery is sufficiently reduced and therefore this result in a very high energy density of the zinc air battery.

- Second, it has a very long shelf life.
- We cannot say unlimited shelf life because there could be some other reasons like corrosion, etc., which can happen when you store the battery for a long time.
- But nevertheless, the shelf life is very long because as long as the battery is kept sealed.
- The air cannot enter.
- So that means oxygen cannot enter the holes which are there on the battery.
- If they are sealed, oxygen cannot enter the battery and therefore the reaction cannot happen.
- Therefore, you can store these batteries for a very long time and therefore they are.
- They have very long shelf life.
- They also do not pose any ecological problems.
- There is a zinc and oxygen which are being used in the battery, and it is a low cost battery because zinc.
- oxygen is anyway taken from directly from the atmosphere and zinc when used in manufacture in mass scale the cost becomes very low so it's a low cost battery.
- And the second important point which we need to know is that along with air, when we open the battery, then there is carbon dioxide also in the atmosphere.
- And since we have used alkali here as the electrolyte, what the electrolyte is KOH, therefore, there could be a reaction between this carbon dioxide and KOH, which is going to result in formation of K2CO3.
- And that is going to block the pathways and therefore reduce the efficiency of this battery.
- You can see the first application is in hearing aids.
- Because these batteries, as I told you just now, can be miniaturized, they can be made into very small sizes.
- So this is about the Zincare battery, and in the next class we are going to look at the batteries which are used profusely in the present day, and that is lithium batteries.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Zinc -air battery transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/029_Zinc_-air_battery__50608371ec91d5a3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zinc -air battery transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/029_Zinc_-air_battery__50608371ec91d5a3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zinc -air battery transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/029_Zinc_-air_battery__50608371ec91d5a3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Zinc -air battery transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/029_Zinc_-air_battery__50608371ec91d5a3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 rendered pages in 034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md have no readable `.slides.md` text; their original page images are included.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.

4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00014	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0044. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0045 to s0064. Sidecar variants compare...
SRC00015	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0044. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0045 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0044. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0045 to s0064. Sidecar variants compare...
SRC00017	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0044. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0045 to s0064. Sidecar variants compare...

C Revision and retrieval practice